PW SKILLS

INTRODUCTION TO

# DATABASE

## DBMS | MySQL | ETL vs ELT | Data Pipeline



# INSTRUCTIONS TO STUDENTS

- Read each question carefully before answering.
- For normalization questions, show each step clearly with table diagrams.
- Partial credit may be awarded for the correct approach with minor errors.
- Attempt ALL questions.

**Q1. What is the difference between a DBMS and a File-Based System? List any THREE advantages of using a DBMS over a file system.**

**Q2. Define the following terms with one example each:**

- Primary Key
- Foreign Key
- Referential Integrity

**Q3. Classify the following data sources as Structured, Semi-Structured, or Unstructured. Give a reason for each:**

- An Excel spreadsheet of employee salaries
- A Twitter/X post
- A JSON response from a weather API
- An MRI scan image

## Q4. You are given the following unnormalized table. Convert it to First Normal Form (1NF). Clearly state the problem with the original table and show the converted table.

Original Table: student_courses

| student_id | student_name | courses_enrolled |
|---|---|---|
| S01 | Alice Kumar | DBMS, Python, Statistics |
| S02 | Ravi Sharma | Machine Learning, SQL |
| S03 | Priya Singh | DBMS, Excel |

## Q5. The table below is in 1NF but violates 2NF. Identify the partial dependency and decompose it into 2NF-compliant tables.

Table: order_items | Primary Key: (order_id, product_id)

| order_id | product_id | quantity | product_name | unit_price | customer_name |
|---|---|---|---|---|---|
| 1001 | P01 | 2 | Laptop | 55000 | Alice Kumar |
| 1001 | P02 | 1 | Mouse | 800 | Alice Kumar |
| 1002 | P01 | 3 | Laptop | 55000 | Ravi Sharma |
| 1003 | P03 | 5 | USB Hub | 1200 | Priya Singh |

**Q6.** **The following table is in 2NF but violates 3NF due to a transitive dependency. Identify the transitive dependency and normalize it to 3NF.**

Table: employee_dept | Primary Key: emp_id

| emp_id | emp_name | dept_id | dept_name | dept_location |
|---|---|---|---|---|
| E001 | Alice Kumar | D01 | Data Engineering | Bangalore |
| E002 | Ravi Sharma | D02 | Analytics | Mumbai |
| E003 | Priya Singh | D01 | Data Engineering | Bangalore |
| E004 | Karan Mehta | D03 | ML Research | Hyderabad |

**Q7.** **Explain the three types of relationships in a relational database with a real-world example for each. For the One-to-Many relationship, draw the table structure showing the Foreign Key placement.**

**Q8.** **What is the difference between ETL and ELT? In what real-world scenario would you prefer ELT over ETL? Support your answer with a specific example.**

**Q9.You are given the following completely unnormalized hospital management table. Normalize it fully through 1NF → 2NF → 3NF. Show each step with the resulting table(s) and state the primary key(s) at each stage.**

Original Unnormalized Table: hospital_data

| adm._id | Patient_name | patient_dob | dr._id | dr._name | dr._dept | treat._given | ward_id | ward_name | ward_floor |
|---|---|---|---|---|---|---|---|---|---|
| A001 | Alice Kumar | 1995-03-12 | D01 | Dr. Mehra | Cardiology | ECG, Blood Test | W01 | ICU | 3rd |
| A002 | Ravi Sharma | 1988-07-25 | D02 | Dr. Kapoor | Neurology | MRI | W02 | General | 1st |
| A003 | Alice Kumar | 1995-03-12 | D01 | Dr. Mehra | Cardiology | ECG, X-Ray | W01 | ICU | 3rd |

**Q10. Design an ER Diagram (describe it in text/table format) for a Library Management System. The system must handle: Books ( book_id, title, publication_year, genre, copies_available), Members(member_id, member_name, email, phone, membership_date), Rentals(rental_id, rental_date, due_date, return_date, fine_amount) and Authors(author_id, author_name, nationality, birth_year). Identify entities, attributes, primary keys, and all relationships with their cardinalities.**

**Q11.** **Analyze the following scenario and answer all parts: A ride-sharing company (like Ola/Uber) processes 2 million trip records daily. Their current ETL pipeline takes 4 hours to complete, but the analytics team needs data available within 15 minutes for real-time driver allocation.**

(a) What is the fundamental problem with the current approach?

(b) Propose an alternative architecture using modern data pipeline concepts.

(c) What tools/technologies would you recommend, and why?

(d) How does your solution ensure data quality?